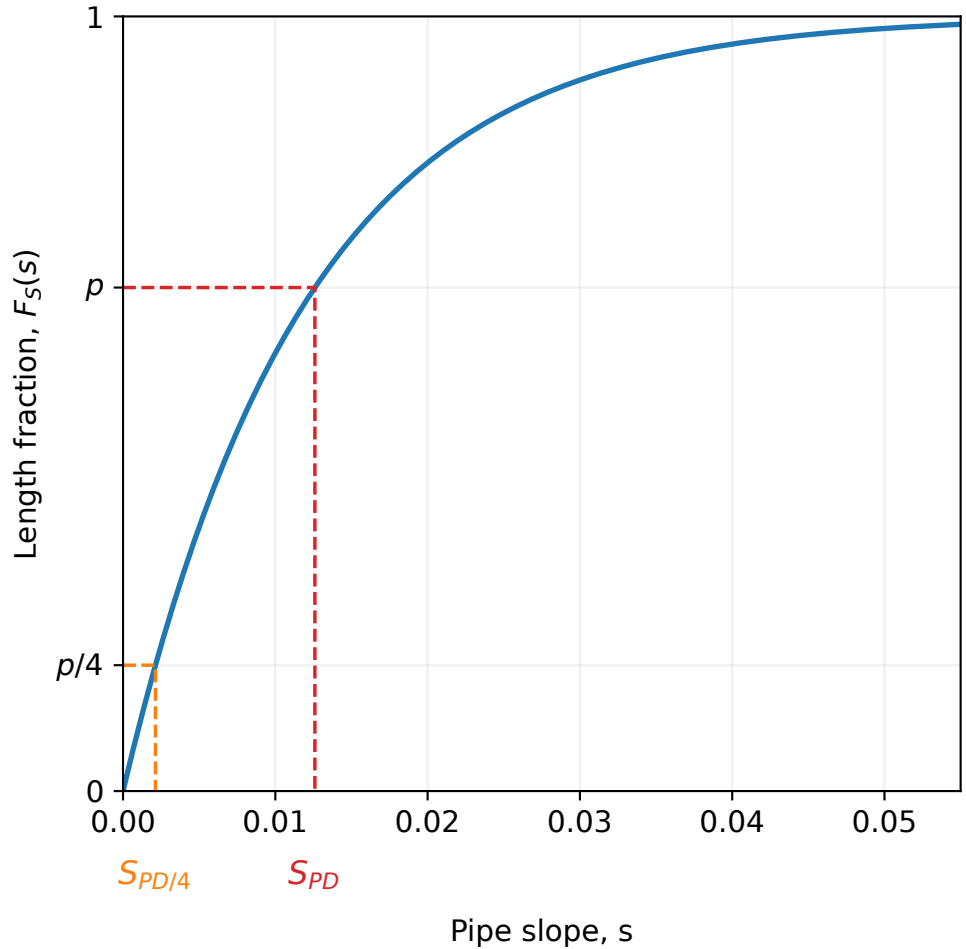
A. Information required by EQ1

Prior deposition-model output
Cumulative deposited mass vs. pipe length

Read length fraction at 80% of total mass
$p = P_{LD}/100$; $L_{PD} = pL$

Combine model-derived p with geometry-derived CDF

Network pipe slopes and segment lengths
Length-weighted cumulative distribution $F_S(s)$

$S_{PD} = F_S^{-1}(p)$; $S_{PD/4} = F_S^{-1}(p/4)$

B. Quantile lookup (illustrative CDF)